Structurally Disadvantaged: A Statistical Analysis of Batting Position and Team Qualification Bias in the IPL Orange Cap Award

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Abstract

The Orange Cap is the most sought-after individual batting award in the Indian Premier League (IPL). In this paper we discuss how it systematically disadvantages middle-order batsmen and high performers who do not qualify for the play-offs. This bias stems from three advantages enjoyed by openers and play-off teams: greater ball-faced volume, exclusive access to power-play conditions, and additional match opportunities from play-off qualification.

Using ball-by-ball data from Cricsheet comprising 278,034 deliveries across 18 seasons, we show that power-play access and play-off matches are strong determinants of the Orange Cap award. We are not comparing players or evaluating player performance; we are assessing the fairness of the award's design.

The results confirm each hypothesis. Openers face a median of 80 more balls per season than middle-order batsmen and score 58% of their runs in the power-play overs. When runs are normalised to a 14-match baseline, 77.6% of the top batters per season rank differently, confirming the advantage of match access. Players who qualify for the play-offs play a median of two additional matches per season, widening the margin for the Orange Cap. In all 18 seasons every Orange Cap winner has batted in positions 1–3.

The findings demonstrate that match access, not batting skill alone, determines the award outcome. The award structurally disincentivises middle-order batting skill. Beyond the unfairness in individual recognition, this has broader implications for the well-documented importance of the number-4 position at the international level. We call for a re-evaluation of the Orange Cap criteria to account for structural positional and match-count advantages.

Keywords: cricket analytics; IPL; T20; batting position; award fairness; sports analytics

1 Introduction

The Indian Premier League is the most popular T20 cricket league in the world (TODO source, 2026). It was established in 2008 and has gained popularity over time, with its total business value reaching $18.5 billion in 2026, fuelled by a $6.4 billion media-rights deal and roughly $600 million in advertising revenue (TODO source, 2023). The league recognises individual performance with several awards at the end of the season, and the most popular of these is the Orange Cap.

To popularise the T20 format, the IPL introduced several individual performance awards to celebrate explosive batting. The Orange Cap is awarded to the batsman who scores the most aggregate runs in the season. Introduced in the inaugural 2008 season, it is a single-criterion award — most runs scored. There is no adjustment for batting position, balls faced, or matches played. The leader gets the privilege of wearing an orange cap during play, and with it comes significant media visibility throughout the season. Ever since its inception, every winner has batted in positions 1–3, with 15 of 18 being openers.

The first six overs of the game are the power-play, where only two fielders are allowed outside the circle, giving the batsman a large advantage in scoring runs. Whether power-play access gives openers an advantage over middle-order batsmen, and whether play-off qualification provides additional opportunities to score, are questions this paper examines empirically.

Prior work on cricket analytics has examined batting position and performance from multiple angles. Pandey and Tolani (2023) used Bayesian survival analysis to study middle-order batsman performance in ODI cricket. Batting role-aware performance indices for T20 cricket have been proposed (Author(s), 2022). Probabilistic approaches to batting-order advantage have also been explored (Author(s), 2020). However, to the best of our knowledge, no prior study has examined whether a major T20 cricket award is structurally biased by batting position and match access.

The remainder of this paper is organised as follows. Section 2 describes the data source and methodology. Section 3 presents results across six analyses. Section 4 discusses the implications of the findings, including the role of the Super Striker award as a partial compensating mechanism. Section 5 concludes with a call for re-evaluation of the Orange Cap criteria.

2 Data and Methodology

[Draft to write. Cover: (i) Cricsheet data source — 1,241 matches, 278,034 deliveries, seasons 2008–2025; (ii) variable definitions — batting position (rank of first appearance per innings), position groups (Opener 1–2, Top Order 3, Middle Order 4–5, Finisher 6+), phase (power-play overs 1–6, middle 7–15, death 16–20), legal balls (wides excluded), play-off team definition; (iii) qualification threshold (7+ matches per season); (iv) statistical tests used — chi-square, Kruskal–Wallis, Mann–Whitney U, Wilcoxon signed-rank, $p < 0.05$.]

Data source

We use publicly available ball-by-ball match data from Cricsheet (Rushe, 2026), covering all IPL matches from the 2008 season through 2025 (18 complete seasons; the ongoing 2026 season is excluded). After restricting to the first two innings of each match and excluding super-overs, the dataset comprises 278,034 deliveries.

Variable definitions

- **Batting position** is the rank of a batter's first appearance in an innings. Per-season position is the average across the batter's innings.
- **Position group** maps average position to one of four groups: Opener (1–2), Top Order (3), Middle Order (4–5), and Finisher (6+).
- **Phase** is derived from the over number: power-play (overs 1–6), middle (7–15), and death (16–20).
- **Balls faced** counts only legal deliveries; wides are excluded, in line with the official IPL strike-rate convention.
- **Play-off team** is any team that appeared in a Qualifier, Eliminator, or Final in a given season.

A batter-season qualifies for population comparisons only if the batter played at least seven matches that season, avoiding single-appearance noise.

Statistical methods

We use the chi-square test for the winner-position distribution (Analysis A); Kruskal–Wallis with Mann–Whitney U post-hoc tests for differences across position groups (Analyses B, C); the Wilcoxon signed-rank test for paired actual-versus- normalised rankings (Analysis D); and one-sided Mann–Whitney U tests for match-count and run differences by play-off status (Analyses E, F). We adopt a significance threshold of $p < 0.05$.

3 Results

[Draft prose to write around each figure/table. The figures, tables and key numbers below are final from the analysis pipeline.]

3.1 Batting position of Orange Cap winners

In all 18 seasons, every Orange Cap winner batted in positions 1–3; 15 of 18 were openers (positions 1–2) and the remaining three batted at position 3 (Figure 1). No middle-order or finishing batsman has ever won. A chi-square test confirms the winner-position distribution differs significantly from the population distribution of qualified run-scorers ($\chi^2 = 49.61$, $p < 0.001$, $df = 3$).

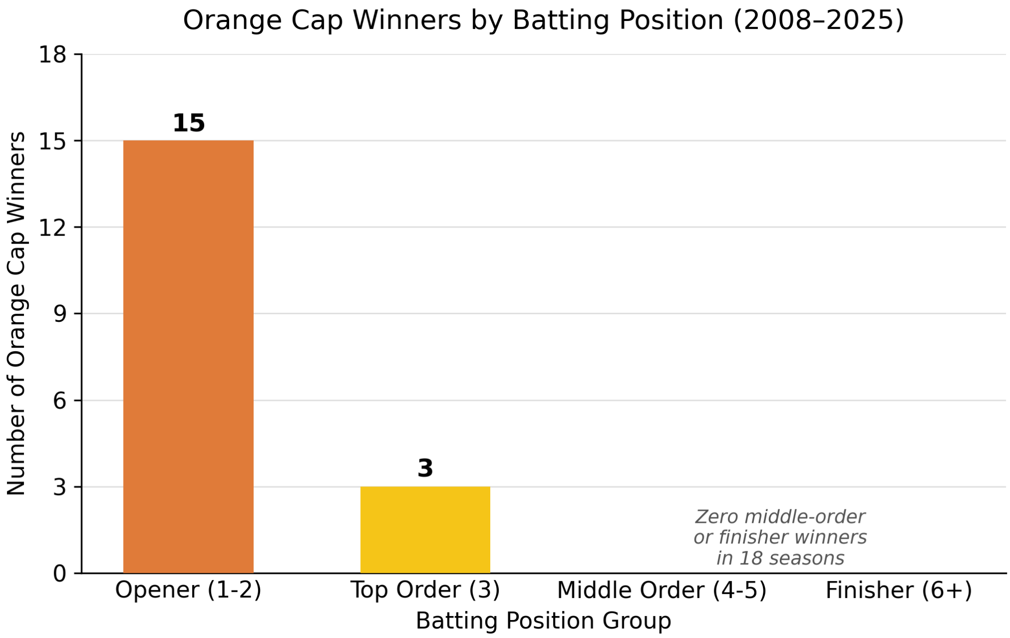

Figure 1: Distribution of Orange Cap winners by batting position group, 2008–2025. Every winner batted in positions 1–3.

3.2 Balls faced across position groups

Openers face a median of 284 legal balls per season, versus 204 for middle-order batsmen — a gap of 80 balls (Table 1, Figure 2). A Kruskal–Wallis test confirms the difference across groups is significant ($H = 1051.8$, $p < 0.001$, $n = 2783$).

Table 1: Balls faced per season by position group (legal balls; 7+ match qualified batter-seasons, 2008–2025).

Position group	Batter-seasons	Median balls	Mean balls	Mean runs
Opener (1–2)	224	284	286.1	386.2
Top Order (3)	183	235	250.7	339.3
Middle Order (4–5)	372	204	209.7	282.4
Finisher (6+)	376	101	112.5	155.8

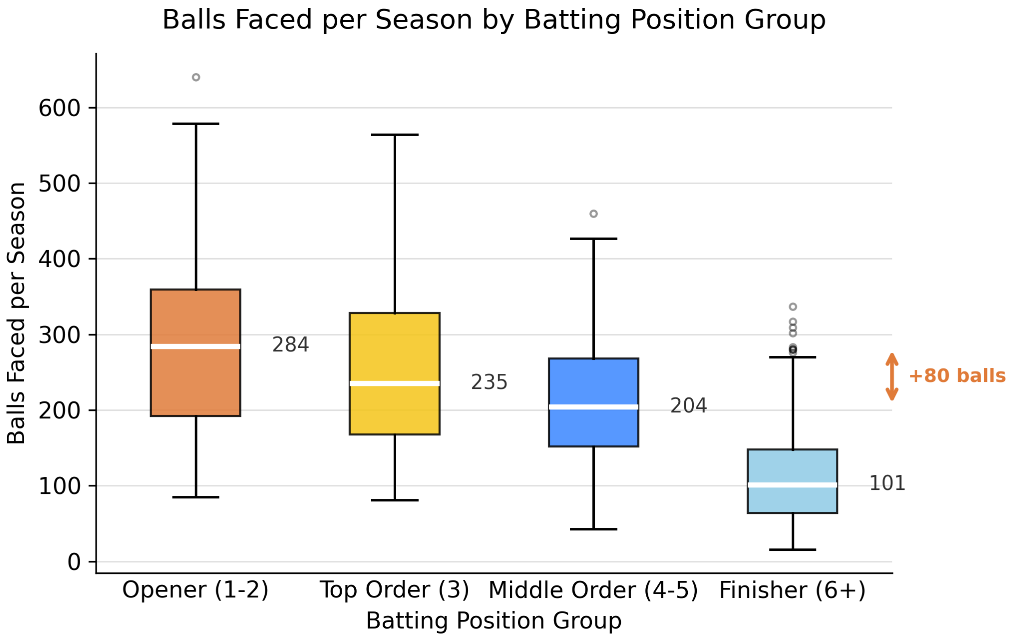

Figure 2: Distribution of balls faced per season by position group.

3.3 Power-play run concentration

Openers score 57.5% of their seasonal runs in the power-play, compared with just 16.2% for middle-order batsmen and 2.4% for finishers (Table 2, Figure 3). The difference in power-play run share across groups is significant (Kruskal–Wallis, $p < 0.001$).

Table 2: Share of seasonal runs by phase and position group (%).

Position group	Power-play	Middle	Death
Opener (1–2)	57.5	35.4	7.1
Top Order (3)	45.1	44.3	10.6
Middle Order (4–5)	16.2	56.4	27.3
Finisher (6+)	2.4	34.6	63.0

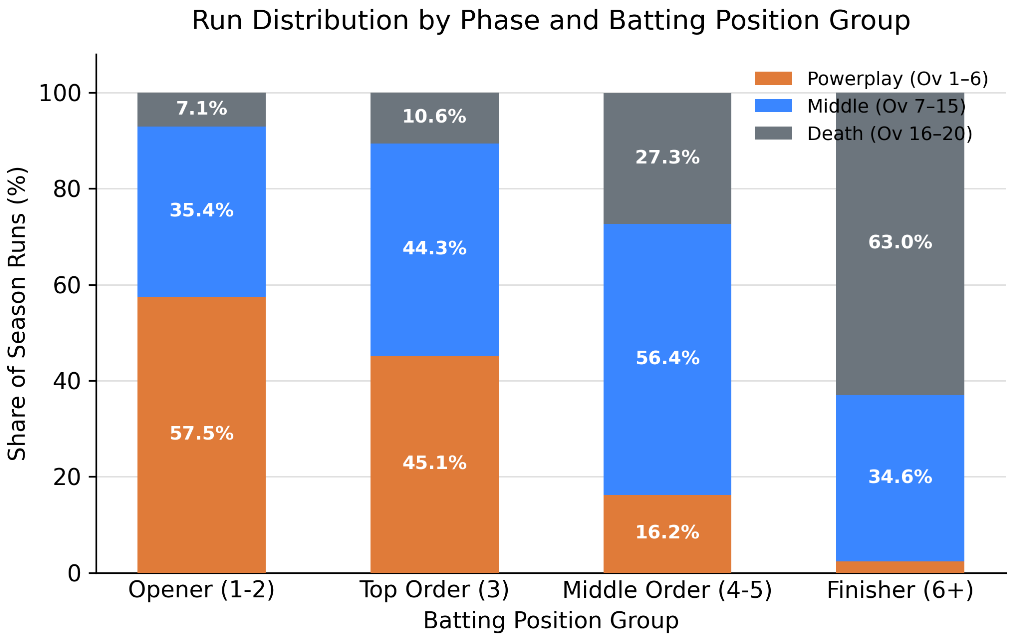

Figure 3: Runs by phase as a share of seasonal total, by position group.

3.4 Runs versus batting position

Across all qualified batter-seasons, seasonal runs correlate negatively with average batting position ($r = -0.61$, $p < 0.001$), as shown in Figure 4.

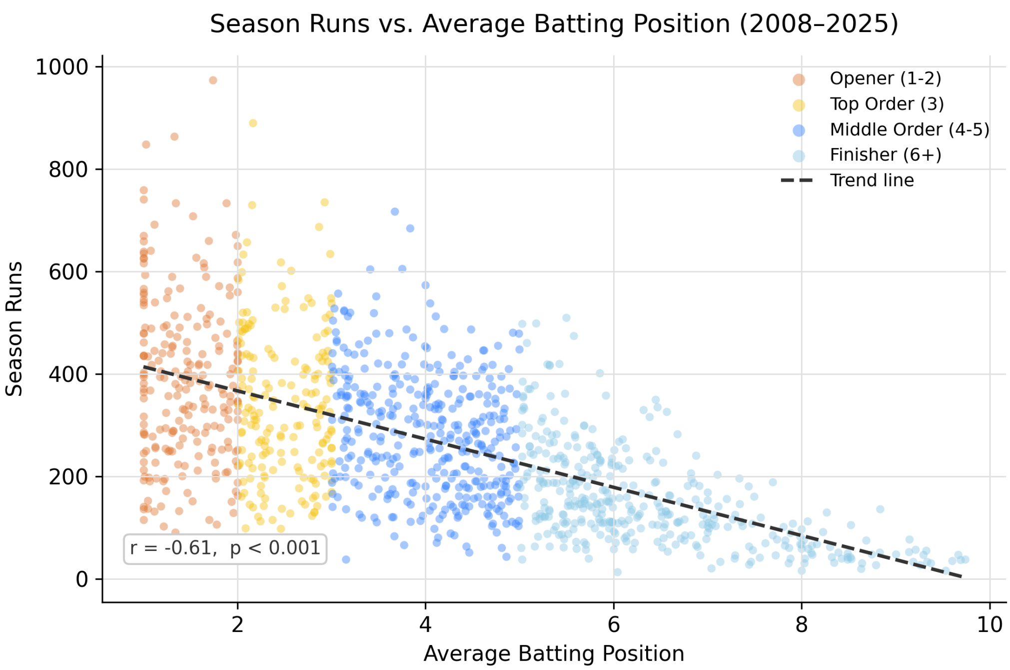

Figure 4: Seasonal runs versus average batting position, all qualified batter-seasons. Negative correlation ($r = -0.61$).

3.5 Normalised-ranking counterfactual

Restricting to league-stage matches and normalising each batter's runs to a common 14-match baseline (norm_runs = league_runs × 14/league_matches), 77.6% of the top-10 rankings per season change order. A Wilcoxon signed-rank test confirms actual and normalised ranks differ significantly ($W = 1914.5$, $p < 0.001$, $n = 183$). Figure 5 shows the ranking shifts; 95% confidence intervals account for batters who played fewer than 14 matches.

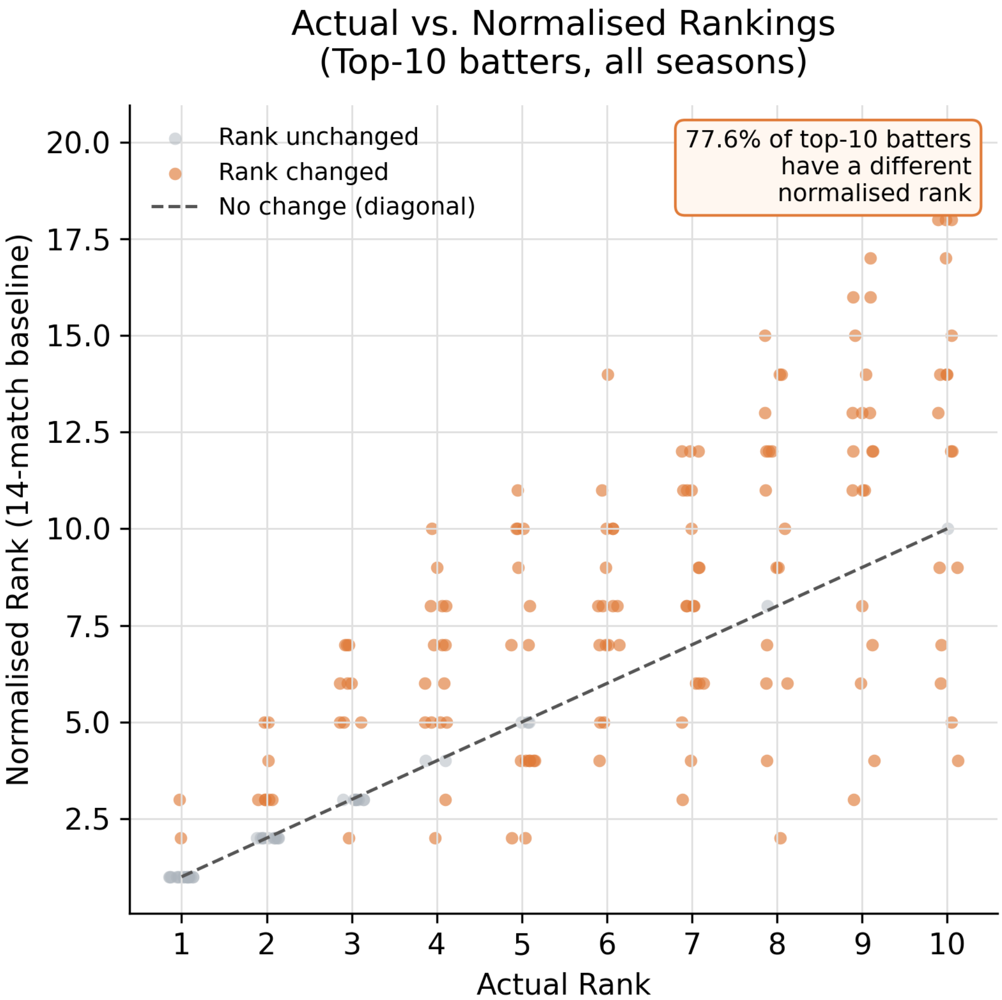

Figure 5: Actual versus normalised rankings of top run-scorers after normalising to a 14-match baseline.

3.6 Play-off match advantage

Play-off teams play a median of two additional matches per season relative to non-play-off teams (Mann–Whitney U, $p < 0.001$; $n_{\text{playoff}} = 72$, $n_{\text{non}} = 84$), as shown in Figure 6. Given a typical opener pace of roughly 40 runs per match, two extra matches translate to a material run cushion in a close Orange Cap race.

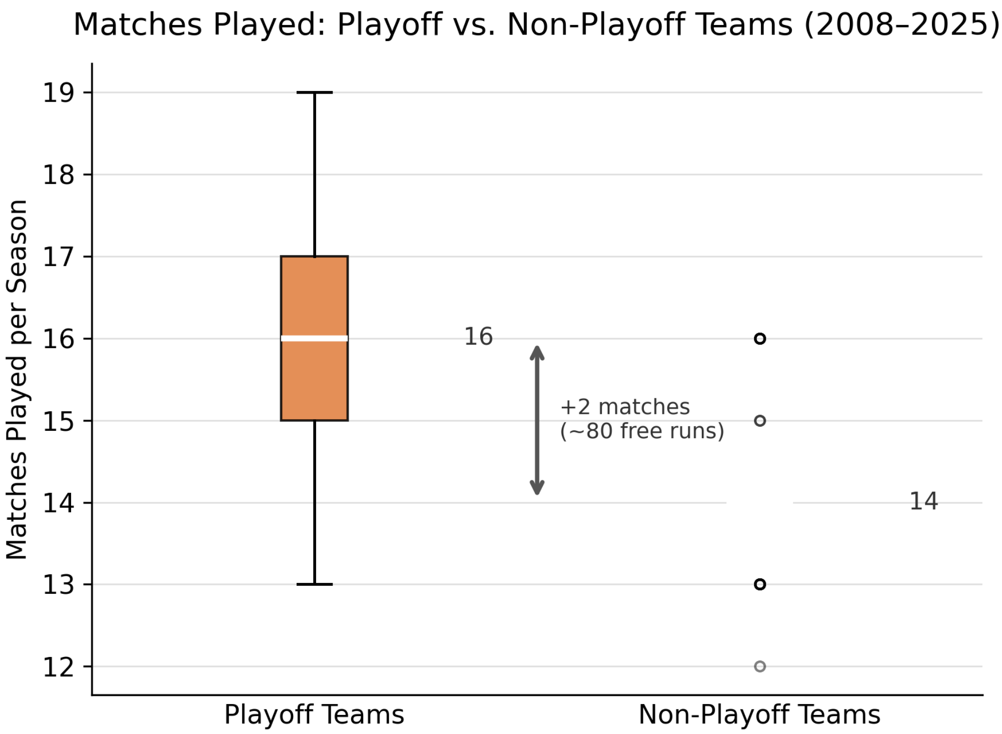

Figure 6: Matches played per season by play-off and non-play-off teams.

3.7 Non-play-off elite batsmen

We identify 23 batter-seasons in which a top-5 run-scorer played for a non-play-off team and was therefore capped at the 14-match league stage. Projecting their runs to a 16-match schedule (the median for play-off teams) shows several would have challenged or exceeded the actual Orange Cap total. Notably, a one-sided Mann–Whitney U test does *not* find that play-off top-5 batters out-score non-play-off top-5 batters ($p = 0.95$; medians 552 vs. 625): elite non-play-off batsmen are not less productive — they simply have fewer matches in which to accumulate runs.

4 Discussion

[Draft to write. Cover: (i) implications for award design — the award rewards volume/access over skill; (ii) the Super Striker award as a partial but inadequate compensating mechanism (introduced 2018, ≥100-ball threshold, last three winners 2024–2026 were openers — converging toward the same pattern); (iii) the broader importance of the number-4 position at international level; (iv) limitations — Impact Player rule (2023+), DLS-shortened matches, opposition quality not modelled, batting position volatility.]

The Super Striker award

The IPL introduced the Super Striker award in 2018, given to the batsman with the highest strike rate (minimum 100 balls faced). Its existence is an implicit acknowledgement that pure run accumulation does not capture batting quality. However, it does not adequately compensate for the Orange Cap bias: it is a secondary award with far less prestige, the 100-ball threshold still excludes many middle-order batsmen, and the last three winners (2024–2026) have all been openers, converging toward the same positional pattern as the Orange Cap itself.

5 Conclusion

[Draft to write. Summarise the structural bias, restate that match access (position + play-off qualification) — not skill alone — determines the award, and call for a re-evaluation of the Orange Cap criteria. Tease Paper 2 (a bias-corrected Best Batsman Index).]

References

Author(s). A probabilistic approach to identifying run scoring advantage in the order of playing cricket, 2020. arXiv preprint. TODO: complete arXiv id and authors.

Author(s). A new in-form and role-based Deep Player Performance Index for player evaluation in T20 cricket. *Decision Analytics Journal*, 4:100–110, 2022. TODO: complete authors, volume, pages.

Mukesh Pandey and Harshit Tolani. A Bayesian perspective of middle-batting position in ODI cricket. *Journal of Sports Analytics*, 9(1):1–17, 2023. doi: 10.3233/JSA-220642.

Stephen Rushe. Cricsheet: Ball-by-ball data for cricket matches. `https://cricsheet.org`, 2026. Accessed: 2026-06-01.

TODO source. IPL media rights deal, 2023. TODO: cite media-rights source.

TODO source. IPL brand and business valuation, 2026. TODO: cite valuation / media-rights source.